

EJERCICIOS DE SQUID

Squid es un servidor proxy

◆ 1. Verificar instalación de Squid

Este ejercicio lo haremos individual, y después en grupo, de manera que el equipo se conectará al proxy de uno de los compañeros.

En Kali ya suele venir en los repositorios. Comprueba:

squid -v

Si no está instalado:

sudo apt update

sudo apt install squid -y

```
jcsantos at parrot in ~  
└─$ sudo squid -v  
Squid Cache: Version 5.7  
Service Name: squid  
Debian linux  
configure options: '--build=x86_64-linux-gnu' '--prefix=/usr' '--includedir=${prefix}/include' '--mandir=${prefix}/share/man'  
 '--infodir=${prefix}/share/info' '--sysconfdir=/etc' '--localstatedir=/var' '--disable-option-checking' '--disable-silent-rul  
es' '--libdir=${prefix}/lib/x86_64-linux-gnu' '--runstatedir=/run' '--disable-maintainer-mode' '--disable-dependency-tracking'  
 'BUILDCXXFLAGS=-g -O2 -ffile-prefix-map=/build/reproducible-path/squid-5.7=. -fstack-protector-strong -Wformat -Werror=format  
-security -Wno-error=deprecated-declarations -Wdate-time -D_FORTIFY_SOURCE=2 -Wl,-z,relro -Wl,-z,now ' 'BUILDCXX=g++' '--with-  
build-environment=default' '--enable-build-info=Debian linux' '--datadir=/usr/share/squid' '--sysconfdir=/etc/squid' '--libexe  
cdir=/usr/lib/squid' '--mandir=/usr/share/man' '--enable-inline' '--disable-arch-native' '--enable-async-io=1' '--enable-store
```

◆ 2. Iniciar y habilitar el servicio

sudo systemctl start squid

sudo systemctl enable squid

sudo systemctl status squid

Debe quedar en estado **active (running)**.

```
jcsantos at parrot in ~  
└─$ sudo systemctl status squid.service  
● squid.service - Squid Web Proxy Server  
   Loaded: loaded (/lib/systemd/system/squid.service; enabled; preset: enabled)  
   Active: active (running) since Tue 2026-01-27 09:42:23 CET; 2min 41s ago  
     Docs: man:squid(8)  
  Process: 5775 ExecStartPre=/usr/sbin/squid --foreground -z (code=exited, status=0/SUCCESS)  
    Main PID: 5778 (squid)  
       Tasks: 4 (limit: 9386)  
      Memory: 17.0M  
         CPU: 170ms  
   CGroup: /system.slice/squid.service  
           └─5778 /usr/sbin/squid --foreground -sYC  
             └─5782 "(squid-1)" --kid squid-1 --foreground -sYC  
               └─5783 "(logfile-daemon)" /var/log/squid/access.log  
                 └─5784 "(pinger)"
```

◆ 3. Configuración básica del proxy

El archivo principal es:

Sudo nano /etc/squid/squid.conf

En este fichero puedes ajustar:

- **Puerto de escucha (por defecto 3128):**

http_port 3128

- **Permitir acceso desde tu red local (ejemplo 192.168.56.0/24 en VirtualBox):**

acl red_local src 192.168.0.0/24

http_access allow red_local

http_access deny all

```
jcsantos at parrot in /etc/squid
└─○ ll
total 348K
drwxr-xr-x 1 root root  88 Jan 27 09:51 .
drwxr-xr-x 1 root root 5.7K Jan 27 09:42 ..
drwxr-xr-x 1 root root  22 Jan 27 09:42 conf.d
-rw-r--r-- 1 root root 1.8K Nov 14 20:44 errorpage.css
-rw-r--r-- 1 root root 175 Jan 27 09:51 squid.conf
-rw-r--r-- 1 root root 335K Nov 14 20:44 squid.conf.back
jcsantos at parrot in /etc/squid
└─○ cat squid.conf
http_port 3128
#Permitir acceso desde tu red local (ejemplo 192.168.56.0/24 en VirtualBox):
acl red_local src 192.168.0.0/24
http_access allow red_local
http_access deny all

jcsantos at parrot in /etc/squid
└─○
```

- **(Opcional) Definir cache/logs:**
Ya vienen preconfigurados en /var/spool/squid y /var/log/squid/.

Después de cambios:

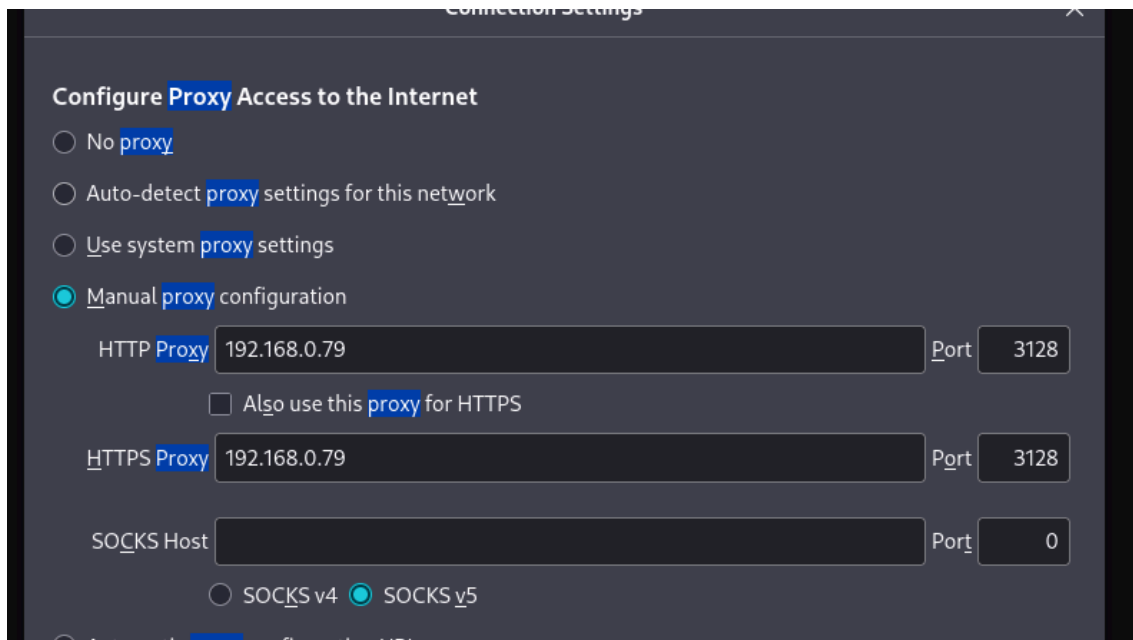
sudo systemctl restart squid

◆ 4. Configurar el cliente (para “usar” el proxy)

Ahora, cualquier cliente (otro Linux, Windows, navegador, etc.) debe apuntar a **la IP de tu máquina virtual + puerto 3128**.

Ejemplos:

- En navegador (Firefox/Chrome):
 - Configuración → Proxy manual →
 - HTTP Proxy: 192.168.56.101 (IP de la VM Kali)
 - Puerto: 3128
 - Usar este proxy para todo protocolo.



- En Linux (temporal):

```
export http_proxy="http://192.168.56.101:3128"
```

```
export https_proxy="http://192.168.56.101:3128"
```

- En Windows (red → configuración de proxy manual):
 - **Servidor proxy: 192.168.56.101:3128.**

*(Asegúrate de que la red de tu VM está en modo **Bridged** o **Host-only** para que el cliente pueda verla).*

◆ 5. Ver logs de uso

Cada vez que un cliente use el proxy, quedará registrado:

`tail -f /var/log/squid/access.log`

```
cat /etc/squid/access.log: Permission denied
jcsantos at parrot in /var/log/squid
sudo cat access.log
1769504193.250 65 192.168.0.108 TCP_TUNNEL/200 2974 CONNECT one.one.one.one:443 - HIER_DIRECT/1.1.1.1 -
1769504194.221 42 192.168.0.108 TCP_TUNNEL/200 2972 CONNECT one.one.one.one:443 - HIER_DIRECT/1.1.1.1 -
1769504327.249 14260 192.168.0.108 TCP_TUNNEL/200 39617 CONNECT www.perplexity.ai:443 - HIER_DIRECT/104.18.26.48 -
1769504327.249 19308 192.168.0.108 TCP_TUNNEL/200 3779 CONNECT www.perplexity.ai:443 - HIER_DIRECT/104.18.26.48 -
1769504327.249 13848 192.168.0.108 TCP_TUNNEL/200 10722 CONNECT r2cdn.perplexity.ai:443 - HIER_DIRECT/104.18.27.48 -
1769504417.340 36 192.168.0.97 TCP_TUNNEL/200 39 CONNECT www.gstatic.com:443 - HIER_DIRECT/142.251.142.131 -
1769504418.036 142 192.168.0.97 TCP_TUNNEL/200 6633 CONNECT ssl.gstatic.com:443 - HIER_DIRECT/142.250.184.3 -
1769504418.041 148 192.168.0.97 TCP_TUNNEL/200 6876 CONNECT ssl.gstatic.com:443 - HIER_DIRECT/142.250.184.3 -
```

◆ 6. Probar funcionamiento

En el cliente, abre un navegador con proxy activado y entra en una web.

Si funciona:

- La página cargará **a través del proxy Squid**.
- En los logs de Kali (/var/log/squid/access.log) verás la petición registrada.

Nota

Probamos a bloquear Google

The image shows a terminal window on the left and a web browser window on the right. The terminal window displays the contents of the `/etc/squid/squid.conf` file, where ACLs are defined to block access to Google domains (google.com, google.es, google) and a rule is set to deny access to these domains. The browser window shows a Firefox error page titled "The proxy server is refusing connections" with the error code "403 Forbidden". The error message states: "Firefox is configured to use a proxy server that is refusing connections. Error code: 403 Forbidden". Below the error message, there are two bullet points: "Check the proxy settings to make sure that they are correct." and "Contact your network administrator to make sure the proxy server is working." A "Try Again" button is visible at the bottom right of the error page.

```
File Edit View Search Terminal Help
jcsantos at parrot in /etc/squid
sudo cat squid.conf
http_port 3128
#Permitir acceso desde tu red local (ejemplo 192.168.56.0/24 en VirtualBox):
acl red_local src 192.168.0.0/24

# ACLs para bloquear Google
acl bloqueo-google-com dstdomain .google.com
acl bloqueo-google-es dstdomain .google.es
acl bloqueo-google-wild dstdomain .google.

# Regla de denegación
http_access deny bloqueo-google-com
http_access deny bloqueo-google-es
http_access deny bloqueo-google-wild

#Regla de permiso a la red local
http_access allow red_local
#http_access deny all
jcsantos at parrot in /etc/squid
```

The browser window shows a Firefox error page titled "The proxy server is refusing connections". The error message states: "Firefox is configured to use a proxy server that is refusing connections. Error code: 403 Forbidden". Below the error message, there are two bullet points: "Check the proxy settings to make sure that they are correct." and "Contact your network administrator to make sure the proxy server is working." A "Try Again" button is visible at the bottom right of the error page.